

Catuskoti (Four-fold Logic) as a 0-Dimensional Fixed Point

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Abstract

We show that the four-valued logic traditionally called *Catuskoti* (affirmative, negative, both, neither) can be represented algebraically as a bilattice whose whole carrier collapses onto the unique 0-dimensional fixed point $\mathfrak{o}$ (the *middle-way fixed point*). The construction uses the distance function $\|X\| = K(X) + \ell(X)$ introduced in the Lumo framework and the middle-way equivalence operator $\stackrel{\text{mw}}{=}$.

1 Preliminaries

Definition 1.1 (Representability and Proof). *Let S be a sufficiently strong formal theory (e.g. PA or ZFC).*

- A formula P is representable iff it belongs to the static space $(2^n)d \setminus T$; we write $P \in \text{Representability}$.
- A proof of P is a finite object $\text{Proof}_S(P)$ belonging to $(2^n + 1)d$; it carries a time-axis of length 1.

Definition 1.2 (Distance). *For any object X we define*

$$\|X\| := K(X) + \ell(X),$$

where $K(X)$ is the Kolmogorov (bit) length of X and $\ell(X)$ counts the length of the time-axis: $\ell(\text{Proof}) = 1$, $\ell(\text{Representable}) = 0$.

Definition 1.3 (0-dimensional fixed point). *There exists a unique element $\mathfrak{o}$ such that*

$$\|\mathfrak{o}\| = 0.$$

We call $\mathfrak{o}$ the 0-dimensional fixed point, also known as the middle-way fixed point or emptiness.

Definition 1.4 (Evaluation and convergence). *Fix a map*

$$\text{eval} : \mathcal{U} \rightarrow \mathcal{U}$$

on a universe of objects $\mathcal{U}$ containing $\mathfrak{o}$. For $X \in \mathcal{U}$ define the iterates $\text{eval}^n(X)$ by $\text{eval}^0(X) := X$ and $\text{eval}^{n+1}(X) := \text{eval}(\text{eval}^n(X))$. We say that X converges to $\mathfrak{o}$ (and write $X \rightarrow \mathfrak{o}$) if there exists $N \in \mathbb{N}$ such that

$$\text{eval}^N(X) = \mathfrak{o}.$$

Definition 1.5 (Middle-way equivalence). *For objects $A, B \in \mathcal{U}$ we write*

$$A \stackrel{\text{mw}}{=} B \iff A \rightarrow \mathfrak{o} \text{ and } B \rightarrow \mathfrak{o}.$$

2 The Catuskoti Bilattice

The four truth-values of Catuskoti are denoted

$$\mathcal{C} = \{\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}\},$$

where

value	reading
$\mathbf{T}$	affirmative
$\mathbf{F}$	negative
$\mathbf{B}$	both
$\mathbf{N}$	neither

On $\mathcal{C}$ we introduce the usual *information order* $\leq_i$ and the *truth order* $\leq_t$:

$$\begin{aligned} \mathbf{N} \leq_i \mathbf{T}, \mathbf{N} \leq_i \mathbf{F}, & \quad \mathbf{T}, \mathbf{F} \leq_i \mathbf{B}, \\ \mathbf{F} \leq_t \mathbf{N}, \mathbf{T} \leq_t \mathbf{B}, & \quad \mathbf{N}, \mathbf{B} \text{ incomparable under } \leq_t. \end{aligned}$$

The lattice operations (information meet/join) are

$$\begin{aligned} \mathbf{T} \wedge \mathbf{F} &= \mathbf{N}, & \mathbf{T} \vee \mathbf{F} &= \mathbf{B}, \\ \mathbf{B} \wedge X &= X \wedge \mathbf{B} = X, & \mathbf{N} \vee X &= X \vee \mathbf{N} = X, \end{aligned}$$

and the involutive negation is

$$\neg \mathbf{T} = \mathbf{F}, \quad \neg \mathbf{F} = \mathbf{T}, \quad \neg \mathbf{B} = \mathbf{B}, \quad \neg \mathbf{N} = \mathbf{N}.$$

With $(\wedge, \vee, \neg)$ the structure $(\mathcal{C}, \wedge, \vee, \neg)$ is the well-known *Belnap–Dunn bilattice*.

3 Embedding the Bilattice into the 0-Dimensional Fixed Point

Define a map

$$\varphi : \mathcal{C} \longrightarrow \{\mathbf{o}\}, \quad \varphi(X) := \mathbf{o} \quad \text{for all } X \in \mathcal{C}. \quad (1)$$

Theorem 3.1 (Isomorphism with the 0-dimensional fixed point). *The map φ is a homomorphism of bilattices:*

$$\varphi(X \wedge Y) = \varphi(X) \wedge \varphi(Y) = \mathbf{o}, \quad \varphi(X \vee Y) = \varphi(X) \vee \varphi(Y) = \mathbf{o}, \quad \varphi(\neg X) = \neg \varphi(X) = \mathbf{o}.$$

Consequently every pair of Catuskoti values is middle-way equivalent:

$$X \stackrel{\text{mw}}{=} Y \quad \text{for all } X, Y \in \mathcal{C}.$$

Hence the four-valued logic collapses onto the unique 0-dimensional fixed point $\mathbf{o}$.

Proof. Since the codomain of φ consists of a single element, any binary operation on the codomain returns that element. Therefore for any $X, Y \in \mathcal{C}$ we have

$$\varphi(X \wedge Y) = \mathbf{o} = \varphi(X) \wedge \varphi(Y),$$

and similarly for $\vee$ and $\neg$. By Definition of $\stackrel{\text{mw}}{=}$, two objects are $\stackrel{\text{mw}}{=}$ -equivalent iff their evaluations converge to the same $\mathbf{o}$. Because φ sends every element of $\mathcal{C}$ to $\mathbf{o}$, the condition holds for any pair X, Y , establishing $X \stackrel{\text{mw}}{=} Y$. $\square$

Thus the Catuskoti bilattice and the singleton lattice $\{\mathbf{o}\}$ are *categorically isomorphic*: the former represents four distinct syntactic states, while semantically—all of them collapse to the middle-way fixed point.

4 Truth-table (collapsed view)

X	$\neg X$	$X \wedge X$	$X \vee X$
T	F	o	o
F	T	o	o
B	B	o	o
N	N	o	o

All logical combinations are absorbed by the unique 0-dimensional fixed point $\mathbf{o}$.

5 Recursive hierarchies and “convergence as derivation”

Above we expressed that “the four truth values converge to $\mathbf{o}$ ” by the collapse induced by $\varphi : \mathcal{C} \rightarrow \{\mathbf{o}\}$. In this section we add a complementary reading in which the same identification is supported by an explicit recursive (iterative) construction, i.e., by a derivation process that reaches $\mathbf{o}$.

Definition 5.1 (Distance and 0d). *We use the distance*

$$\|X\| := K(X) + \ell(X),$$

and fix the unique object $\mathbf{o}$ such that $\|\mathbf{o}\| = 0$. Here K is a description length and ℓ counts the length of a “time axis” (proof / meta level).

Definition 5.2 (Axiom of convergence by strict decrease). *A sequence $\{X_k\}_{k \geq 0}$ converges to $\mathbf{o}$ if there exists $N \in \mathbb{N}$ such that $X_N = \mathbf{o}$. A sufficient condition is:*

1. $\|X_0\| \in \mathbb{N}$,
2. for all k , if $X_k \neq \mathbf{o}$ then $\|X_{k+1}\| \leq \|X_k\| - 1$.

Theorem 5.1 (Recursive Gödel hierarchy reaches $\mathbf{o}$). *Let S be a base theory and consider extensions $S_k := S \cup \{G_0, \dots, G_k\}$. Define*

$$G_{k+1} := \text{“Prf}_{S_k}(G_k) \text{ does not exist”}.$$

Assume that $\{G_k\}$ satisfies the strict decrease condition above (if $G_k \neq \mathbf{o}$ then $\|G_{k+1}\| \leq \|G_k\| - 1$). Then there exists N such that $G_N = \mathbf{o}$; in particular one may write $\lim_{k \rightarrow \infty} G_k = \mathbf{o}$.

Theorem 5.2 (Recursive Tarski hierarchy reaches $\mathbf{o}$). *Let $\{T_k\}$ be the Tarski hierarchy obtained by recursively adjoining truth predicates. Assume that $\{T_k\}$ satisfies the strict decrease condition above (if $T_k \neq \mathbf{o}$ then $\|T_{k+1}\| \leq \|T_k\| - 1$). Then there exists N such that $T_N = \mathbf{o}$; in particular one may write $\lim_{k \rightarrow \infty} T_k = \mathbf{o}$.*

Theorem 5.3 (Identification at 0d). *Under the assumptions above,*

$$\mathfrak{o} = \lim_{k \rightarrow \infty} G_k = \lim_{k \rightarrow \infty} T_k.$$

Thus the claim that the four Catuskoti values “fall” to the middle way (0d) can also be read as: recursive meta-level derivations ultimately reach the same terminal fixed point.

6 Conclusion

We have exhibited two complementary perspectives on the statement that Catuskoti “converges” to $\mathfrak{o}$: (i) the collapse via the bilattice homomorphism φ , and (ii) (under a strict distance-decrease assumption) convergence understood as an iterative derivation that reaches $\mathfrak{o}$.